

Instrumentation for Motor Imagery-based Brain Computer Interfaces relying on dry electrodes: a functional analysis

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Abstract—The functional analysis of a novel instrumentation for Brain-Computer Interfaces (BCI) is carried out. This consists of a wireless wearable helmet with only 8 dry electrodes. The brain signals to be measured through an electroencephalography are related to the sensorimotor cortex. The final aim is to distinguish between different motor imagery tasks. Furthermore, this analysis also takes into account the discrimination between two executed movements. Features are extracted from the brain signals by means of a Common Spatial Pattern algorithm. Then, two different classifiers are employed to process the brain signals, namely the Random Forest, and the Support Vector Machine with Gaussian kernel. Their performance was compared in terms of classification accuracy and the best accuracy resulted equal to about 80% when distinguishing between left and right imagined movement, classified by means of the Random Forest. The results of this study aim at giving a contribution to the building of wearable BCIs for daily life applications.

Index Terms—brain-computer interfaces, wearable devices, motor imagery, functional analysis

I. INTRODUCTION

A Brain-Computer Interface, or BCI, is an alternative communication channel, whose operation does not require peripheral nerves and muscles [1]. Instead, this instrument relies on direct measurement of brain activity, which is translated into a command associated to the user's intention. Electroencephalography (EEG) is widely considered to measure the brain electrical activity [2], [3]. Meanwhile, the output command is derived by processing the brain signals with techniques ranging from simple power analysis to modern machine learning algorithms [4]–[6].

Indeed, BCIs have been mainly used as assistive devices for people with motor impairments and in rehabilitation [7]–[9]. However, the application domain for these interfaces has become broader and broader, and they are today exploited not only to replace or restore lost nervous functions, but

also to enhance, supplement, and improve the natural brain pathways, as well as research tools [10]. Despite that, brain-computer interfaces are still little used outside a laboratory [11]. Meanwhile, people and researchers could benefit from the spreading of this technology in daily life. The foreseen implications are mainly two: (i) exploiting the advantages of a new assistive technology in human activities [12]–[14], and (ii) the possibility to measure brain signals extensively in order to enhance the researches on brain.

There are several factors that limit the spreading of BCI technology, such as the successful recognition of the user's intention in an acceptable time [15], the problem of motion artifacts [16], costs, and wearability [11]. For sure, EEG is a measurement technique that mitigates these last issues because it allows the construction of wearable and portable instruments, it is inexpensive and safe, and it also provides centimeters of spatial resolution and milliseconds of temporal resolution [2], [17]. Furthermore, the electroencephalography does not require the penetration of the skull, which is an essential requirement for a wide usability of the final device. Previous researches in our group concerned a wearable non-invasive BCI based on steady-state visually evoked potentials (SSVEP) [8], [18], and the aim was to build a low-cost instrument whose metrological performance was suitable for daily life applications [14], [19]. However, SSVEP signals require visual stimulation, which can be troublesome for some applications. For this reason, the present work takes into account another well-known BCI paradigm, which is the “motor imagery” (MI) [20]–[23]. In this case, the brain signals to be measured are related to the imagination of a movement, without requiring the user to execute it.

MI-based BCIs relying on EEG typically employ many wet electrodes in order to measure brain activity. These wet electrodes require gels to enhance conductivity and hence the quality of the measured signals. As an example, the widespread dataset of BCI competition IV employed 22 Ag/AgCl electrodes in the recording setup [24]. On the other hand, it is clear that conductive gels must be avoided when aiming at building an easily wearable device [22], and also the number of channels should be reduced [25]. In the cited works it is demonstrated that BCIs based on dry electrodes are compatible in terms of functionality with devices exploiting conductive gels, though performance is usually slightly lower.

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Furthermore, the brain signal processing is clearly a crucial aspect [26], [27].

In the present paper, a wearable EEG helmet with only 8 dry electrodes is employed to measure brain signals related to movement. The functionality of the system is evaluated by considering the discrimination of different movement tasks, which are either executed or imagined. In particular, Section II describes the instrument, proposes a processing of brain signals, and introduces the reference to which the functionality of the instrument is compared. Then, Section III describes the experimental setup and reports the results along with their discussion.

II. MATERIALS AND METHODS

The scope of this work is to analyze the functionality of a wearable EEG helmet for measuring brain signals related to movement. The performance of the employed device is compared to the performance obtained by means of the same processing on a widespread dataset, namely the BCI Competition IV dataset 2a [24]. This dataset was created by means of a wearable cap with wet electrodes and, in this work, it is assumed as a reference for the motor imagery measurement. Following, the instrumentation for acquiring EEG signals and the reference dataset are introduced, and then the processing algorithms adopted for brain signals classification are presented.

A. Instrument description

The EEG acquisition system employed in this study is a wearable helmet (*ab medica*[®] *Helmate* [28]), with 24 bit resolution and CMRR equal to -115 dB (Fig. 1). The helmet is *Class IIA* certified according to Medical Device Regulation (UE) 2017/745, and guarantees 8 acquisition channels (unipolar configuration), through 10 dry electrodes [29], [30]. These are made of conductive rubber and their endings are coated with Ag/AgCl. The electrodes different shapes are designed to pass through the hair and reach the skin as shown in Fig. 2. In principle, all the three types are used depending on the user's scalp. In particular, the shapes (a) and (b) of Fig. 2 are typically used in the central and occipital areas, while the shape (c) is employed in the frontal area because of the hair absence.

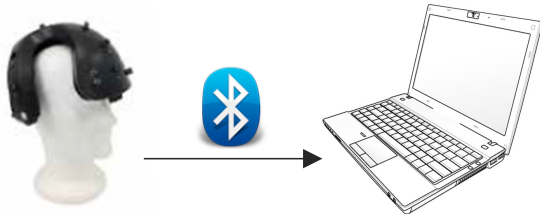


Fig. 1. The wireless wearable helmet connected via Bluetooth to a PC.

According to the international 10-20 EEG system [31], they are located at Fp1, Fp2, Fz, Cz, C3, C4, O1, O2 (Fig. 3). The

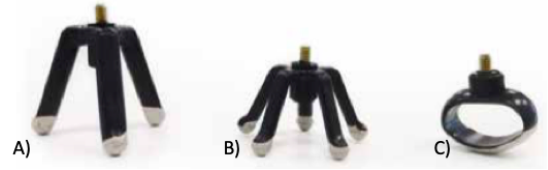


Fig. 2. Electrodes designed to directly touch the scalp (c), even through thick hair (a, b).

ground electrode and the reference electrode are placed in the frontal region at Fpz and AFz, respectively.

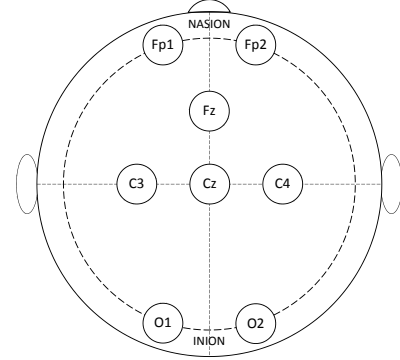


Fig. 3. Position of the dry electrodes in the employed EEG acquisition system.

The signals are recorded as the difference between the i -th electrode and the ground electrode, and the resulting voltage signal is referenced with respect to AFz. These are collected at a sampling rate of 512 Sa/s, and the helmet communicates via Bluetooth with a PC. The contact of the electrodes with the head is checked by using a dedicated software (*Helm8 Software Manager*) which measures the contact impedance. Furthermore, a large variety of filters can be applied to the signals, so that the recorded signal is already pre-processed. In this case, the signal is bandpass-filtered in the 1–45 Hz range and a notch filter at 50 Hz is applied.

B. Reference dataset

As mentioned above, the signals of the BCI Competition IV dataset 2a [24] are considered as a reference for comparison purposes. This dataset comprises EEG signals from 9 subjects, related to 4 classes of motor imagery. Twenty-two Ag/AgCl electrodes with conductive gel were used to record the EEG. All signals were recorded with the left mastoid serving as reference and the right mastoid as ground. The signals were sampled at 250 Sa/s and bandpass-filtered between 0.5 Hz and 100.0 Hz. An additional 50 Hz notch filter was applied.

C. Signal processing

In order to distinguish between different tasks, two machine learning algorithms are adopted in this work for signal processing. Both algorithms consist of two steps, features extraction and classification (Fig. 4).

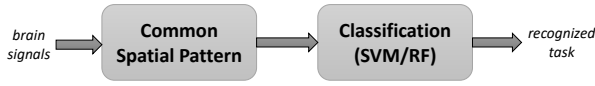


Fig. 4. Brain signals processing for the recognition of the task related to movement: features extraction with Common Spatial Pattern (CSP) and classification (SVM vs RF).

For the features extraction, a Common Spatial Pattern (CSP) algorithm is used to enhance the signal-to-noise ratio of the EEG epochs [32]. It rearranges the data by using the eigenvectors associated with the m largest and m smallest eigenvalues. The resulting features are the input of the classifier. In the classification step, two different classifiers are taken into account. They are described hereafter.

1) *Support Vector Machine (SVM)*: the idea behind this classifier is to find a hyperplane that guarantees the best separation between data points of different classes while maximizing the margin [33]. In order to deal with an eventual non-linear separability of data, the input of the classifier is mapped to a features space with higher dimension through a “kernel function”. Usually, a “Gaussian kernel” is employed. This is also a suggested kernel for the analysis of non-stationary signals like EEG signals. Errors are allowed during separation, and the minimization of the separation error is the key for the training of this classifier.

2) *Random forest*: this classifier consists of a decision tree that uses a bootstrap aggregation (bagging) ensemble procedure developed by Breiman [34]. A random forest is a collection of decision trees, each one considering a different random sample of the input. Hence, each tree is grown using a different random subset of the predictor variables to determine the binary splits. The final decision is made by considering the majority among the decisions of the trees. Usually, if p is the total number of predictors, about $\sqrt{p}$ variables are selected for each tree. The advantages in using this classifier are the low number of requested testing samples to achieve good performance, if compared to other classification algorithms. Nonetheless, it is not easy to interpret, and hence control, every aspect of the trees net [35].

III. EXPERIMENTAL VALIDATION

In this section, the results of a preliminary experimental campaign are reported and discussed. The experimental setup is described first, in order to clarify the conditions in which the system was tested. Then, the results allow to retrieve some information about the functionality of the helmet and address future researches about the possibility to employ few dry electrodes to build a motor imagery-based BCI.

A. Setup

Two subjects (a male and a female, both 26 years old) were enrolled in this preliminary study. Both of them had normal vision and no history of central nervous system abnormalities. The subjects were seated in a comfortable chair during the experiments in a quiet room, and the BCI paradigm consisted

of different movement tasks, either executed or imagined. The experimental protocol consisted of two sessions, carried out in two different days. The data from the first session are only related to the movement execution tasks, while the data from the second session are also related to motor imagery. Each session comprised 5 different exercises: squeeze a soft ball, dorsiflexion of the ankle, flex-extension of the forearm, finger mobilization by clenching a clothespin, and flex-extension of the leg. Every exercise can be executed with the left part of the body or the right part. For instance, one can squeeze the ball with the left hand and the right hand, or, one can move the left or the right ankle.

In the protocol concerning the first session, for each exercise, the BCI user had to execute different trials by alternating 10 s of movement and 10 s of relax. In total 8 movement trials and 8 relax trials are executed for each exercise. The movement trials are divided by alternating a “left movement” and a “right movement”. Furthermore, 30 s of baseline are acquired at the beginning and at the end of the protocol (Fig. 5).

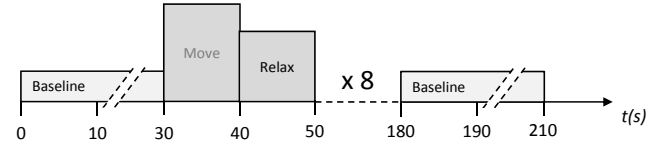


Fig. 5. Experimental protocol of the first session: executed movement and relax tasks.

The protocol concerning the second session, instead, alternates for each exercise 10 s of movement execution, 10 s of motor imagery with closed eyes, and 10 s of relax. In total 8 executed movement trials, 8 imagined movement trials, and 8 relax trials are carried on for each exercise. It is worth noting that motor imagery requires an extensive user training [36].

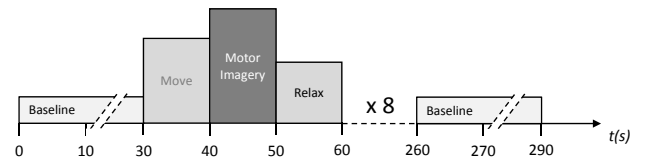


Fig. 6. Experimental protocol of the second session: executed movement, imagined movement, and relax tasks.

B. Experimental results

In the present analysis, the signal processing is conducted by dividing each 10 s trials of the first session into 4 trials which are 2 s long, while the first and last second are discarded. This is done to achieve a higher number of trials, and the 2 s window was chosen as a length compatible with other experimental protocols, e.g. [24]. Thus, in this work, 64 trials are available for each exercise. However, the protocol with 10

s long tasks was conceived to study the EEG signals behaviour in a sustained task, which will be the topic of a further study. Instead, from the second session, only the trials related to the imagined movements are extracted, and again each 10 s trial is divided into 4 trials which are 2 s long. The remaining data will be useful for further studies.

The helmet with dry electrodes allows the acquisition of EEG data through a dedicated software. A .edf file is generated for each exercise of a session. Then, a MATLAB script converts this into a .mat file by separating each trial and associating a label to them. Five different labels are possible, though the discrimination is pairwise: "relax" (R), "executed right movement" (ER), "executed left movement" (EL), "imagined right movement" (IR), and "imagined left movement" (IL). The classification of two tasks is carried out, as described in the previous section, by processing the signals with the CSP filtering and then employing two different classifiers. The quantity employed as a performance indicator is the classification accuracy obtained in a 5-fold cross-validation. The results are reported in the following tables.

The data of each subject are considered separately. The training and validation procedure are repeated 5 times within the cross-validation procedure. Moreover, the five exercises are considered by themselves in the first trials, and all together in the last trial. This allows to retrieve an information about which type of movement is most suitable for the distinction between left and right with this kind of setup. Although each movement is related to a specific brain area, the same channels configuration is adopted for each exercise. In this way the usability of the device is improved for further applications. The extraction of the more informative signal features with respect to each motor task is carried out in the transformed domain after the application of the CSP algorithm.

In Tab. I, the distinction is conducted between a "right executed movement" (ER) and a "left executed movement" (EL).

	Subject 1		Subject 2	
	CSP+RF	CSP+SVM	CSP+RF	CSP+SVM
ex.1	71.7%	60.0%	57.5%	60.8%
ex.2	90.0%	72.5%	61.7%	55.8%
ex.3	72.5%	75.0%	65.0%	57.5%
ex.4	60.0%	55.8%	62.5%	59.2%
ex.5	74.1%	75.0%	70.8%	65.0%
all	70.0%	62.5%	69.4%	54.4%

TABLE I

A RIGHT EXECUTED MOVEMENT VERSUS A LEFT EXECUTED MOVEMENT (ER vs EL).

Instead, in Tab. II, the discrimination is conducted between a "right imagined movement" (IR) and a "left imagined movement" (IL).

These results are discussed in details in the following subsection.

C. Discussion

Some important considerations can be derived from Tab. I and Tab. II. The first is that, as a general trend, the two

	Subject 1		Subject 2	
	CSP+RF	CSP+SVM	CSP+RF	CSP+SVM
ex.1	72.5%	59.2%	65.8%	62.5%
ex.2	65.8%	60.0%	62.5%	36.7%
ex.3	65.0%	74.2%	63.3%	59.2%
ex.4	79.2%	55.8%	59.2%	46.7%
ex.5	79.2%	76.7%	58.3%	49.2%
all	71.9%	63.1%	64.4%	60.0%

TABLE II

A RIGHT IMAGINED MOVEMENT VERSUS A LEFT IMAGINED MOVEMENT (IR vs IL).

classifiers lead to compatible performance, but the Random Forest is usually better. The second important consideration is that the capability of discriminating between executed tasks is compatible with the discrimination between imagined tasks. Next, it can be seen that the performance is subject-dependent and task-dependent. Subject 1 had previous training, both in movement execution and imagination, and its performance results better than subject 2 one, which had no previous training.

Fig. 7 shows the results for the executed movements in a graphic way for the five different exercises. The accuracy related to all the exercises considered together, instead, is not shown. It can be seen that the worst exercise, in terms of accuracy, is for exercise 4 (finger mobilization by clenching a clothespin), while the best exercise is number 2 (dorsiflexion of the ankle) for subject 1, and number 3 and (flex-extension of the forearm and flex-extension of the leg, respectively) for subject 2. It is worth noting that, also for subject 1, exercises 3 and 5 are associated to a good performance. The accuracy for these tasks is about 65% - 70%.

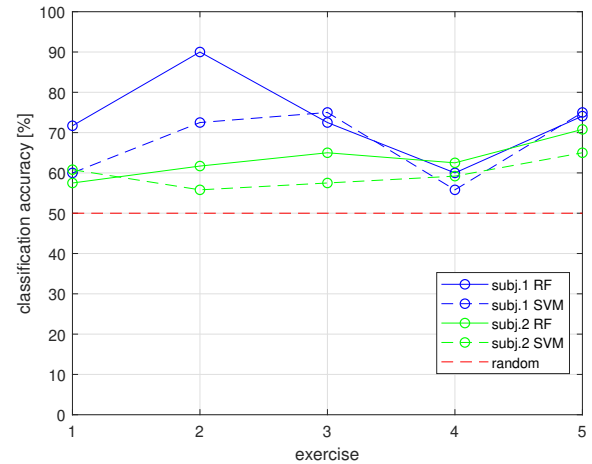


Fig. 7. Classification accuracy for the five exercises with executed movements, considered for the two different subjects and for the two classifiers. The random classification accuracy for two tasks is also shown.

Fig. 8, instead, shows the results for the imagined movements for the five exercises. The overall accuracy trend is decreasing with respect to the executed movements, but some peculiarities are present. It can be seen that the SVM leads to bad performance for subject 2, which was not trained, while

the Random Forest leads to almost the same accuracy value for all the exercises, i.e. around 60% - 65%. Instead, for subject 1, the Random Forest seems again to be preferred, and the best accuracy is reached for the imagination of the finger mobilization (exercise 4) and flex-extension of the leg (exercise 5). This accuracy is almost 80%.

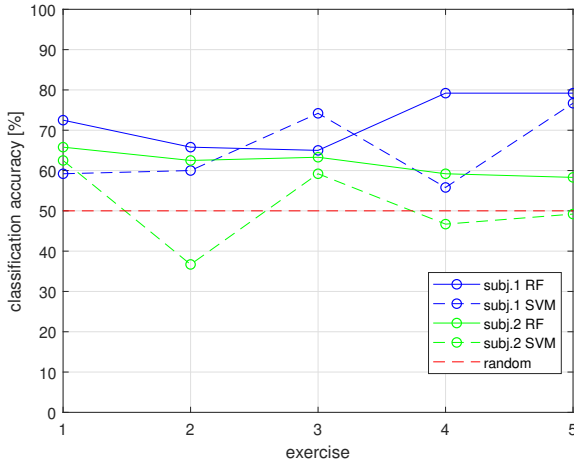


Fig. 8. Classification accuracy for the five exercises with imagined movements, considered for the two different subjects and for the two classifiers. The random classification accuracy for two tasks is also shown.

These results show that the accuracy performance depends both on the exercise and on the subject. They address the need of further investigations, because there is no strong evidence in the scientific literature about the type of movements to prefer. The results also suggest that some motor imagery tasks can lead to better performance with respect to motor execution. This is in accordance with [37], where it is shown that the decoding accuracy of motor execution is compatible with the decoding of motor imagery. Finally, the results are also compatible with the accuracies reached on the reference dataset with the same processing. In particular, in comparing the reported results with the reference dataset ones, only two classes of motor imagery were considered for the latter, i.e. right hand versus left hand. Neither the tongue nor the (both) feet class were considered, because they have no correspondence with the five rehabilitation exercises proposed in this work. Furthermore, though the reference dataset contains data from 22 wet electrodes, only the best 8 were considered. These are the 8 electrodes that allow the best discrimination between the two abovementioned classes, and they were selected with an iterative procedure that aimed to maximize the classification accuracy. Instead, it should be emphasized that it was not possible to directly consider the same 8 electrodes of the helmet because the reference dataset does not include the electrodes Fp1, Fp2, O1 and O2. The 5-fold cross-validation has been adopted for the 9 different subjects of the reference dataset in order to distinguish the imagination of right hand versus left hand with the Random Forest or the SVM. Depending on the subject, the classification

accuracy goes from 58.2% to 90.2%, with an average accuracy equal to 67.5% for the Random Forest and 71.9% for the SVM. Moreover, it is to report that the mean classification accuracy achieved with all the 22 wet electrodes of the reference dataset is about 80%, which is still compatible with the best classification achieved with the 8 dry electrodes of the helmet. Nonetheless, it is clear that more electrodes, which are also wet, lead to better performance.

In conclusion, the helmet analyzed in this work, by only relying on 8 dry electrodes, allows a data acquisition which is suitable for the needs of a motor imagery-based BCI. However, further investigations are needed in order to better understand which the achievable performance with such a wearable non-invasive instrument is.

IV. CONCLUSIONS

In this paper, the functional analysis of a wireless wearable helmet relying on 8 dry electrodes has been discussed. This novel instrumentation allows the construction of a non-invasive motor imagery-based BCI, which is suitable for daily life applications. The analysis has been conducted by evaluating the classification accuracy performance of two distinct classifiers, the Random Forest and the Support Vector Machine, employed for the processing of data acquired with the helmet after a Common Spatial Pattern filtering was applied to them.

With some preliminary results, it has been shown that the achieved classification accuracy depends both on the subject and on the movement task, either executed or imagined. This accuracy goes up to 80%, and it is also compatible with the classification accuracy obtained, in the same conditions, on a reference dataset employing wet electrodes.

The results of this work suggest that the instrumentation here taken into account can be employed for the decoding of motor imagery, even though some more questions are left open for further research and development.

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